Research statement

Ivan Sabolić

It is becoming obvious that the next decade will be defined by AI and its impact on society, for better or for worse. Experiments that would have taken weeks to run last year can now be delegated to autonomous agents to receive results within a few hours. I expect that most of my technical work, as well as most of the practical machine learning research, will be automated very soon. However, making sure this transition goes well is a defining challenge for the field and a prerequisite for **responsible deployment at scale**. Although frontier models capture immense amounts of knowledge, their behavior remains difficult to anticipate and explain, posing safety and security risks. That raises core questions about how to exercise **control and monitoring**; how to build **resilient safeguards**; and how to design **evaluations and red-teaming** that predict real-world performance.

During the first part of my PhD, I focused on defenses against **data-poisoning backdoors** in computer vision. Drawing on core machine-learning principles, I developed two backdoor-robust training methods. The first defense [1] leveraged generative modeling of self-supervised representations to detect suspicious examples and train a robust model. The second approach [2] addressed the same threat using variational inference with the EM algorithm. Both approaches reached state-of-the-art performance in defending image-classification task against backdoor attacks. However, motivated by some great talks on Adversarial ML [3], I've shifted my research focus to the **robustness and alignment of Large Language Models**. I am currently wrapping up a project on a backdoor defense for the Large Vision Language models, which will soon be available as a preprint. This work proposes a modified training objective for **robust instruction-tuning** on poisoned data that upper bounds population risk on unavailable clean data. In sum, these approaches push the state of the art in backdoor robustness while offering the combination of empirical performance, theoretical footing, and engineering practicality required for real-world adoption.

With this foundation, I aim to develop reliable, scalable LLM systems that encompass **control, safeguards, and high-fidelity evaluation**. The poisoning and backdoor threats I have studied in computer vision now extend directly to large language models [4], and monitoring deployments for such attacks raises its own practical challenges [5]. Also, the evaluation landscape itself is increasingly fragile: adversarial ML problems are growing harder not only to solve but also to evaluate reliably [6], and widely-adopted safety procedures such as machine unlearning have been shown to lack conclusive evaluations [7]. Without **trustworthy evaluation**, neither alignment-oriented [8] nor control-oriented approaches [9] to AI safety can be validated in practice.

I want to visit Prof. Abdelnabi because her work on **contextual integrity** may connect naturally to my own. Frontier models are no longer single-turn assistants. For instance, Claude Opus 4.6 supports **Agent Teams** where multiple LLM instances exchange messages, claim shared tasks, and broadcast findings to one another. In such architectures, governing what information flows to whom is not just a policy question but a model capability, and contextual integrity [10] offers the right framework for it. What interests me most is the adversarial angle: poisoned training data could corrupt a model's contextual integrity, and adversarial prompts could bypass them entirely. My background in robust training applies directly here, and while my previous work has theoretical groundings, I enjoy the empirical side of ML research as well. Running experiments on scale allows us to understand how these systems actually behave. Working with Prof. Abdelnabi in Tübingen feels like the right place to do that.

References

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